

This question tested candidates' knowledge and understanding on work, energy and power using the example of a lift. Candidates' performance was satisfactory. Part (a) was well answered.

In (b)(i), only the more able candidates worked out correctly the total mechanical power output of the motor when the counterweight was installed. Most knew that in (b)(ii) the output power required from the motor was smaller with the addition of the counterweight. Few candidates were able to explain clearly why the claim in (b)(iii) was not true. Some wrongly thought that the lift could still be raised even when the drum's surface was frictionless.

Part (a) was in general well answered. Many candidates attempted to tackle part (b) in terms of current passing through the voltmeter rather than the voltage across AB. As a result, many failed to explain precisely how the reading was affected. In (c)(i), for those who were able to find the resistance of R by considering the potential difference across the resistors, most managed to work out the correct answer. Some candidates did not attempt part (c)(ii) which involved unfamiliar situation. For those who did attempt the question, not many were able to explain the action of the circuit.

This question was on the projectile motion of an acrobat. Candidates' performance was satisfactory. In (a), some candidates tried to solve the problem by using an equation of motion instead of conservation of energy to find v_B . Most candidates were able to apply correct equations to find the answers to (b)(i)(ii) although a few got v_x , v_y and v_B mixed up.

This question was well answered. In (a), a significant number of candidates misunderstood the phrase 'per km' and therefore wrongly divided the resistance by 1000 m. Some confused surface area with cross-sectional area and thus failed to find the correct answer. In (a)(iii), many candidates did not realise that electric shock is essentially due to the large amount of current passing through the body but not merely because of high voltage. Few were able to mention that negligible current would pass through the bird's body. Quite a number of them wrongly thought that the cable was covered with insulating material. Answers like 'The potential difference across the bird is zero' or 'There is no current passing through the bird's body' were common. Part (b) was well answered. A few candidates forgot to multiply the resistance per km ($0.05 \cdot \text{km}^{-1}$) by 10 km in (b)(ii). In (b)(iii), the terms stated by some candidates were not precise, for example, 'cable' was used instead of 'wire', 'sliced iron' was used instead of 'laminated core' etc.

This question tested candidates' understanding of the loading effect of meters in simple circuits. Candidates' performance in (a) was satisfactory. In (a)(i), some candidates took the product of the total current and the resistance of the resistor concerned as the voltmeter reading. Most candidates knew that the finite internal resistance of the voltmeter introduced error to the measurement but few understood how to work out the problem mathematically. Some candidates in (a)(ii) failed to state explicitly that the resistance of the voltmeter should be 'much higher' than that of the part under measurement. In (b)(i), many candidates wrongly held that both voltmeter and ammeter readings are not true values for the resistor concerned. Not many obtained the correct percentage error in R in (b)(ii).

This question tested candidates' knowledge and understanding on projectiles. In general, candidates' performance was fair. Most answered part (a)(i) correctly. However, some failed to distinguish the horizontal uniform motion and the vertical uniformly accelerated motion of a projectile in (a)(ii). Weaker ones even employed $v^2 = u^2 + 2as$ to compute the resultant speed of the projected bearing. Part (b) revealed that quite a number of the candidates had misconceptions about Newton's third law of motion - taking weight and air resistance as an action-and-reaction pair in this case.

This question tested candidates' understanding of the relationship among voltage, current and resistance in a circuit. Candidates' performance was poor. Part (a) was well answered. In (b), few were able to state explicitly that the temperature increase led to the increase in resistance. Not many pointed out that by definition $R = V/I$ in part (c). Although most were able to obtain the correct resistances in (d), not many realised that the one corresponding to room temperature should be employed to find the length of the tungsten filament in part (e).

This question tested candidates' knowledge and understanding in Mechanics in the context of life nets. The overall performance was fair. Part (a)(i) was well answered. Many candidates omitted the falling person's weight in finding the average force acting on the person by the net in (a)(ii). In (b)(i), some candidates failed to point out the possible consequences for a person falling from a great height. Not many realised that in (b)(ii) the determining factors for reaching the life net's central part are the initial horizontal velocity and the time of fall of the projectile motion, not air resistance or wind, which were neglected.

Part (a) was in general well answered. Most candidates were able to draw the circuit diagram correctly though some failed to mark the polarities of the voltmeter or wrongly connected it across the switch. Quite a number of the candidates suggested precautions that were not relevant to accuracy or safety of the experiment like 'using a voltmeter (assumed ideal) of a very high resistance'. In (b), not many gave a correct expression of V in terms of ϵ , r and R . A significant number of them mistook that V varied linearly with R or they had a directly proportional relationship.

Candidates' performance was satisfactory. Some candidates wrongly arranged the speeds in the opposite order in (a). In (b), a lot of the candidates confused the following: acceleration, weight, acceleration due to gravity and velocity. In (c)(i), most were able to describe the energy conversion correctly. However, some failed to point out that the energy conversion from B to C only involved 'part of the sphere's kinetic energy'. Part (c)(ii) was well answered. Candidates were able to use energy conservation to solve the problem. About half of them managed to tackle (c)(iii) by resolving the components of the initial velocity of the sphere. Some even tried to work out the time of flight separately based on the independence of horizontal and vertical motions.

Candidates' performance was fair. Although many were able to indicate the forces in the free-body diagrams in (a)(i), they did not pay attention to the points of application of the forces concerned. In (a)(ii), some candidates did not explicitly point out the directions of the net force acting on the hanging mass and the resulting acceleration. Weaker ones held that there was acceleration even though they stated that the forces were balanced. Parts (a)(iii) and (b) were quite challenging. Candidates had to realise that the total mass of the system was constant (i.e. $(M + 0.1)$ kg) throughout the experiment. Thus, only the more able ones obtained the correct equation of a straight line in (a)(iii) and found M correctly in (b).

Candidates' performance was fair in general. Some candidates failed to relate the average thrust to the rate of change of momentum of the exhaust gas in (a)(i). In applying Newton's laws of motion to find the rocket's acceleration in (a)(ii), weaker candidates omitted the weight mg or wrongly took the net force as $F_{\text{thrust}} + mg$. Candidates performed poorly in (a)(iii). Not many were able to point out that the average thrust remained the same while the rocket's mass decreased. A few candidates mistook the period of the satellite around the Earth as 1 year or 365 days in (b)(i). In (b)(ii), only the more able ones knew how to make use of $GM/R^2 = g$ to tackle this part.

Candidates' performance was satisfactory. Many candidates realised that resistors R and S were in parallel and thus answered (a)(i)(ii) correctly. Weaker candidates made mistakes in the units of currents or mistook 9 V to be the p.d. across the two parallel branches. Parts (a)(iii)(iv) were well answered. In (b), very few knew the purpose of introducing a protective resistor R in the branch with an ammeter. Some wrongly stated that the ammeter reading would be infinite without R, or a short circuit would occur if R was connected in parallel with the ammeter.

The overall performance was satisfactory. Most candidates were able to find the gymnast's kinetic energy in (a) although a few gave a wrong unit 'N' for energy. In (b)(i), not many got full marks in describing the conversion of mechanical energy of the gymnast to the elastic potential energy of the trampoline. Part (b)(ii) was poorly answered. Not many were able to employ the principle of energy conservation and work done on the trampoline to estimate the average force exerted by the gymnast. Some failed to obtain the correct answer as they omitted the further loss in gravitational potential energy of the gymnast after the contact was made.

Candidates' performance was fair. Most were able to comprehend the circuit in (a) for finding the I-V characteristics of the light bulb. Many candidates correctly calculated the bulb's rated resistance in (b)(i). In (b)(ii), most knew that the bulb's resistance increased with the applied voltage but few pointed out that there was an increase in current/power. Candidates did poorly in (c). Many did not realise that in the circuit concerned the bulb was not working at its rated values. Thus even for those who correctly drew the required straight line on the I-V characteristics graph, not many were able to utilize the pair of current and voltage readings to find the power in (c)(ii).

The overall performance of candidates was fair. Part (a) was well answered although a few candidates confused resistance and resistivity. Most answered correctly in (b)(i). Candidates did poorly in (b)(ii) as many of them failed to identify the three branches connected in parallel across MN, namely, R (2.0Ω), CF (0.5Ω) and $CDEF$ (1.5Ω). More than half of the candidates understood that the potential difference across R was greater in (b)(iii). Not many realised that in (b)(iv) there was no potential difference across the coil with S connected to terminal 1, thus the power dissipated should be zero. Some candidates knew that there was no current in the coil but they wrongly related this to 'short circuit'.

This question tested candidates' understanding of simple circuits in the context of a kind of deformation sensors known as strain gauge. The overall performance was satisfactory. Candidates knew that the metal foil's resistance increases in (a), however, many overlooked that the cross-sectional area A of the foil decreased when the foil was stretched. (b)(i) was well answered while some candidates were not careful enough in substituting the resistance values into the given equation in (b)(ii). Less than half of the candidates made a correct deduction in (b)(iii)(II).

This question assessed candidates' application of kinematic concepts to projectile motion. The overall performance was fair. Part (a) was generally well handled, though some only calculated the time to reach the highest point. Others overlooked the opposing directions of initial velocity and acceleration. In (b), a number of candidates were apparently unaware of the horizontal velocity at the highest point and incorrectly claimed that the kinetic energy there is zero. Responses to (c) were poor; most did not mention that the horizontal velocity of the two objects is identical, and many mistakenly argued that the two objects travel the same distance.

This question examined candidates' understanding of circuit behaviour and interpretation of experimental data. The overall performance was good. Most answered (a) correctly. In (b), a high percentage calculated the total resistance accurately, though some only noted the difference (0.3 Ω) without explaining its cause.

This question tested candidates' understanding of the efficiency of energy conversion and the power loss in long-distance transmission of electricity. Candidates' performance was satisfactory. Part (a) was well answered, with a very high proportion obtaining the correct answer in (a)(i), and about two thirds completing the calculations in (a)(ii) and (iii). A small number made errors with prefixes of the units. Part (b)(i) was well answered. In (b)(ii), just under half determined the power loss correctly. The performance in (b)(iii) was poor. Some omitted the use of a transformer, while others failed to explain the benefit of high-voltage transmission.

(No Section B candidate-performance note.)

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